Loud Music Causes Hearing Loss

It is not uncommon among concertgoers to leave a concert unable to hear. In most cases their hearing returns shortly. Unfortunately, permanent hearing loss from loud concerts is becoming increasingly common. The situation can be even more serious for musicians, because they suffer more frequent and prolonged exposure to loud music than most people. It can also be more devastating for a musician to lose his hearing, because his livelihood and passion often leave with his hearing.

Sound levels at concerts can be in the range of 120 to 140 db, well beyond the 100 db normally recognized as the threshold at which short-duration exposure can cause hearing loss. The loss is caused by damage to fragile tissue strands within the cochlea. These strands are called hair cells and resemble tiny hairs. They move with the fluid in the cochlea to stimulate the electrical impulses in the auditory nerve. The hair cells become damaged in the presence of loud noise.

The mechanism of damage is much like the mechanism by which constant walking on grass can damage it. A person walking on a lawn occasionally doesn't damage the lawn, because the grass has a chance to recover between tramplings. But if the grass is subjected to frequent traffic, it soon loses its ability to spring back and becomes permanently damaged. Damage to the hair cells occurs the same way. The louder and more frequent the exposure to loud noise, the more damage the hair cells sustain.

Fortunately, our ears often warn us when we have subjected them to potentially damaging sound levels. Temporary hearing loss is one indication; ringing in the ears is another. If you experience either of these symptoms after exposure to loud sounds, it's an almost sure indication of temporary damage that could become permanent with repeated exposure.

For people who are unable or unwilling to remove themselves from noisy situations, earplugs can help preserve hearing. Earplugs are especially effective at reducing the intensity of high frequency sound, which is what does the most damage to the hair cells. A 15 or 20 db reduction in the intensity of high frequency sound can delay or prevent hair cell damage.